A Pharmacokinetics Evaluation of a New, Low-Volume, Oral Sulfate Colon Cleansing Preparation in Patients With Renal or Hepatic Impairment and Healthy Volunteers

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The pharmacokinetics (PK) of an oral sulfate solution (OSS) for bowel cleansing preparation was studied. OSS (30 g of sulfate) was split between 2 doses, 12 hours apart. Safety measures included electrocardiography, vital signs, adverse events, hematology, blood chemistry, and urinalysis. Six adult patients with moderate renal disease (MRD), 6 with mild-moderate hepatic disease (M/MHD), and 6 normal healthy volunteers (NHVs) completed the study. Adverse events were mild to moderate in severity and were mainly limited to headache and expected gastrointestinal symptoms. Serum sulfate levels were highly variable at all times, even after adjusting for baseline. Sulfate was higher in MRD in comparison to the other groups. The C_{max} and AUC were higher in the patients, but no statistically significant differences emerged. Sulfate levels returned to predose values within 54 hours after dosing. No electrolyte disturbances occurred. Urinary sulfate excretion was approximately 20% of the dose. OSS was well tolerated. The types and severity of adverse events were similar to those seen in large phase III trials. While patients with MRD had elevated sulfate, the levels were less than those in renal failure and did not alter biochemical parameters that are associated with hypersulfatemia.

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The present approaches to bowel cleansing for colonoscopy have significant drawbacks. The large-volume (up to 4 L) isotonic solutions, while not causing clinically significant fluid or electrolyte shifts, are sometimes difficult for patients to ingest.[1,2] Concentrated oral sodium phosphate (OSP) solutions or tablets require an indeterminate amount of supplemental water.[2] In addition to being contraindicated in patients with clinically significant renal impairment, renal failure occasionally ensues from calcium phosphate deposition in the renal tubules, even in patients without identified risk factors. This "acute phosphate nephropathy" is an emerging issue in association with OSP.[3,4] The United States Food and Drug Administration (FDA) recently recommended that over-the-counter (OTC) OSP products should not be labeled or used for bowel preparation and required that prescription OSP products carry a "boxed warning" on the risks of phosphate nephropathy.[5] Thus, there is a medical need for a safe and effective small-volume orally administered colonic purgative formulation that avoids these problems.

In comparison to OSP, administration of a colonic purgative formulation of oral sulfate salts (OSS) has a diminished propensity to precipitate urinary calcium.[6] Large, controlled clinical studies have evaluated this oral sulfate solution (OSS) for its safety and effectiveness to cleanse the colon in preparation for colonoscopy.[7]

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METHODS

Study Design

This was an open-label, single center, in-clinic study of an OSS formulation that was administered as 2 doses separated by 12 hours. Safety measures, pharmacokinetics (PK), and clinical chemistry evaluations were assessed. This trial was registered at www.clinicaltrials.gov as NCT00583.

Study Population

Adult male and female normal healthy volunteers (NHVs) and patients with mild or moderate hepatic impairment (M/MHD, Child-Pugh stage A or B) or moderate renal impairment (MRD, glomerular filtration rate [GFR] of 30-60 mL/min/1.73 m^2) participated in this trial. Half of the NHV group was matched by age, body mass, gender, and race to the hepatic patients and half to the renal patients. Excluded were subjects who tested positive for illegal drugs, or had clinically significant gastrointestinal disease, uncontrolled electrolyte abnormalities, severe renal or hepatic insufficiency, congestive heart failure, or impaired consciousness that predisposed to aspiration.

Study Center

DaVita Clinical Research (Minneapolis, Minnesota) conducted the clinical portions of this study. An independent Institutional Review Board approved the protocol, amendments, and the informed consent form. All subjects granted their written informed consent before any study procedures. This study was in full compliance with federal regulations (including the ICH/GCP Guidelines and HIPAA Privacy Regulations).

Study Medication

The OSS (SuPrep, Braintree Laboratories Inc) consisted of sodium sulfate (35.0 g), potassium sulfate (6.3 g), magnesium sulfate (3.2 g), and flavoring agents, divided into two 6-oz liquid doses. It provides approximately 30 g of sulfate, identical to the formulation studied in Phase III.[7] The study staff diluted each dose with water into 16 oz and supervised its consumption at 6 AM and at 6 PM on day 1. The subjects also drank 2 additional 16-oz cups of water 1 to 3 hours after each dose.

Study Procedures

Subjects who met entry criteria at a screening visit ate a normal breakfast and lunch on the day before admission (day –1). They returned to the Clinical Research Unit (CRU) by 2 PM on day –1 to revalidate entry criteria and stayed until 12 noon on day 2. After a light dinner before 8 PM, they fasted, consuming only clear liquids ad libitum until 8 PM on day 1 when they had a standard dinner. A standard breakfast (before 8 AM) and lunch (before 12 noon) were available on day 2. Subjects left the CRU at 12 noon on day 2. A physical examination included the measurement of weight, vital signs, and 12-lead electrocardiogram before 12 noon on days 1 and 2. They returned to the CRU before 12 noon on the mornings of days 3 and 6 for the collection of blood and urine.

Blood samples were collected approximately 10 minutes before dose 1 and at 1, 2, 4, 8, and 10 hours thereafter and at approximately 10 minutes before dose 2 and at 1, 2, 4, 8, 12, and 18 hours thereafter. Hematology was performed at screening, before dose 1 on day 1, and at 12 noon on day 2. Urine was collected prior to dose 1 (a single void) and then over 0 to 6, 6 to 12, 12 to 24, and 24 to 30 hours. Blood and urine samples were also collected before 12 noon on days 3 and 6. MedTox Laboratory (Minneapolis, Minnesota) performed all routine analyses, while Mayo Medical Laboratories (Rochester, Minnesota) analyzed urinary sulfate.

WIL Research Laboratories LLC (Ashland, Ohio) developed and validated an assay for serum sulfate via ion chromatography (IC) using a Dionex IonPac AG11 guard column (50 × 4.0 mm) and a Dionex IonPac AS11 analytical column (250 × 4.0 mm) (Dionex Corporation, Sunnyvale, California). The conductivity at 35°C (with a suppressor current of 189 mA) allowed calculation of the samples' concentrations from the unweighted ln quadratic function derived from the theoretical concentrations of calibration standards added to blank human serum. The method was validated over the sample concentration range of 50 to 200 ppm sulfate using a 200 µL sample and had an intersession relative standard deviation of <4.0%.

Statistical Methods

The results from all subjects who took OSS (the intent-to-treat [ITT] population) formed the basis for analysis in this study. All statistical analyses were performed using SAS version 8.2 or higher (Chicago, Illinois). The estimation of PK parameters was

Table I Mean (±SD) Urinary Sulfate and Serum Pharmacokinetics Parameters

Parameter	MRD (N = 6)	M/MHD (N = 6)	NHV (N = 6)
Urinary concentration (mmol/dL)			
Predose 1	948 (417)	906 (521)	1366 (489)
Day 3	1410 (899)	935 (935)	1516 (1146)
Day 6	1009 (495)	737 (870)	1401 (778)
Cum. % dose$_{(0\text{-}30)}$	17 (5)	22 (5)	20.3 (13)
Serum PK parameter			
C_{max}, μmol/L	717 (270)	560 (153)	499 (165)
$AUC_{(0\text{-}\tau)}$, mmol*h/L	12 (4)	11 (3)	8 (3)
T_{max}, h	17 (3)	14 (5)	17 (8)

SD, standard deviation; cum. % dose $_{(0\text{-}30)}$, cumulative percentage of dose excreted up to 30 hours; C_{max}, maximum observed concentration; $AUC_{(0\text{-}\tau)}$, area under the curve for the 24-hour dosing interval; T_{max}, time to maximum concentration after dose 1.

performed using PK Solutions 2.0, NonCompartmental Pharmacokinetics Data Analysis, Summit Research Services (Montrose, Colorado). Values for all missing data points remained so, with no imputation. Burt Seibert, PhD (StatNet, Plaistow, New Hampshire), completed the PK assessments and the statistical comparisons.

Electrolytes and Safety

One-way analysis of variance (ANOVA) assessed the differences between the NHV group with each of the MRD and M/MHD groups for clinical chemistry, hematology, urinalysis, and PK. All adverse events were analyzed based on the principle of treatment emergence.

Analysis of PK Variables

Table I defines the urinary and PK parameters that were computed. Actual sampling times were used in the analysis. The serum PK parameters were calculated from the baseline-adjusted serum concentrations. Any resulting negative serum concentrations were set to an arbitrary value of 1.0. Urine sulfate parameters were tested for differences between the NHV group and each of the patient groups using a Wilcoxon test statistic.

RESULTS

Disposition and Demographics of Subjects

Twenty-nine subjects were screened, of which 18 were enrolled and completed the study with 6 patients in each group. Illicit drug use and abnormal clinical chemistry findings were the primary reasons for exclusion. Six adult patients with MRD (3 females), 6 with M/MHD (4 females), and 6 NHVs (4 females) completed the study. Their ages (39-66 years) and weights (121-189 lb) were similar across the groups. However, the number of blacks and whites differed among the groups, representing 1 and 4, 5 and 1, and 4 and 2 for each race for the MRD, M/MHD, and NHV groups, respectively. Most (5/6) of the patients with hepatic impairment had class A Child-Pugh scores (5-6 points), while one had moderate impairment (class B; 8 points). Hepatitis C was the primary disease associated with hepatic impairment in 5 of 6 patients; alcoholic cirrhosis contributed to the other case. The 6 patients with renal disease had moderate renal impairment according to their GFRs (42-48 mL/min) before dose 1. There were no significant protocol violations that rendered any of the patients' data to be ineligible for analyses.

Safety

All patients were 100% compliant with medication consumption. One subject experienced a serious adverse event, elevated creatine kinase (CK), prior to receiving study medication and was withdrawn from the study. There were no deaths or severe adverse events. No patients withdrew from the study because of a treatment-emergent adverse event. Four patients (22%) had no adverse events, and 14 (78%) experienced 24 adverse events. The adverse events were distributed among the groups' health status in a similar fashion: 7 for the NHVs, 8 among the M/MHDs, and 9 for MRDs. Of the events, 19 were judged mild, the rest were considered moderate, and all resolved without sequelae. Headache and gastrointestinal symptoms (abdominal cramps, nausea, and emesis) were the most frequent and commonly occur with bowel-cleansing regimens. Ten other adverse events (including 2 patients with elevated serum creatinine and symptomatic hypoglycemia) were all considered mild, not clinically significant, and resolved by day 6. No patient with renal disease had a worsening of his or her creatinine levels.

Vital Signs and Other Observations

No clinically significant abnormality in vital signs occurred. Some subjects showed abnormal electrocardiogram results at baseline and/or postdose (mostly sinus bradycardia and voltage criteria for left ventricular hypertrophy), which the investigator deemed not clinically significant.

Serum Chemistry, Hematology, and Urinalysis

The investigator considered all serum chemistry, hematology, and urinalysis results not clinically significant and/or consistent with the underlying condition in the case of the patients. There were no differences in serum sodium, potassium, calcium, bicarbonate, and magnesium or anion gap between the patient groups and the healthy volunteers. Any other differences between the patient groups and the NHVs were evident before and sometimes after dosing and were not considered related to OSS as a result.

Sulfate PK

As expected, all subjects had detectable sulfate levels before dosing. Figure 1 shows that predose levels (unadjusted for baseline) in NHVs and patients with M/MHD were similar and mostly within the range (240-420 µmol/L) generally found by IC methods in healthy people.[8] As expected, patients with MRD had higher sulfate levels before dosing and at all times thereafter compared to NHVs and those with M/MHD. Sulfate concentrations generally increased in most patients within 1 hour after dosing, decreased after reaching a peak 2 to 4 hours postdose 1, but did not return to predose levels before dose 2. After dose 2, mean sulfate levels climbed above those after dose 1, increased until T_{max} (2-4 hours postdose 2), and began declining thereafter, returning to predose ranges by day 3 or 6. There were no statistically significant differences in the ln weight–adjusted PK parameters for sulfate between each of the patient groups and the healthy volunteers (Table I).

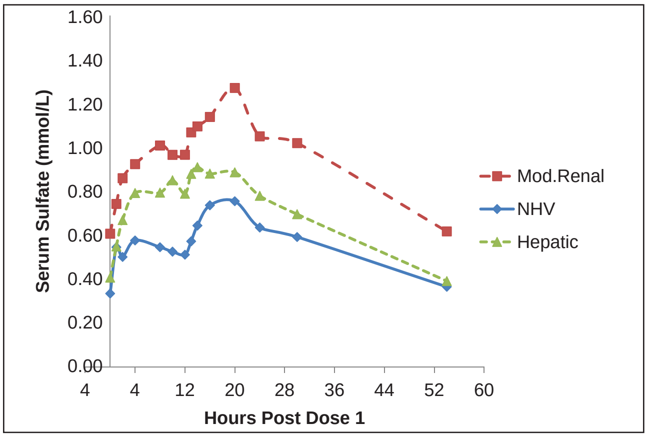

Figure 1. Serum sulfate levels (Mean ± SE) after dose 1 (0 hours) and dose 2 (12 hours) of oral sulfate solution. MRD, moderate renal disease patients; M/MHD, mild-moderate hepatic dysfunction patients; NHV, normal healthy volunteers.

Urine Sulfate

Urine sulfate parameters varied widely among individuals (Table I). The cumulative amount of urinary sulfate excretion over 30 hours after dose 1 was similar among the groups. There were no differences with regard to the urine volume or the percentage of the sulfate dose that was excreted over the 30-hour period. Assuming that OSS was the only sulfate source, approximately 17% to 22% of the administered dose appeared in urine over this period. The true fraction of the excreted dose is considerably smaller because this calculation does not consider basal sulfate excretion.

DISCUSSION

This study evaluated an OSS intended for bowel cleansing on sulfate PK, safety measures, and clinical chemistry in patients with MRD, M/MHD, and healthy volunteers who were not undergoing colonoscopy. To our knowledge, this is the first detailed examination of the PK of sulfates in humans. OSS transiently elevated serum sulfate in all groups, especially in those with MRD. Comparing these elevations to those in other disease states places them in clinical context. For example, in renal failure, hypersulfatemia is symptomatic at 2000 to 8000 µmol/L, considerably higher than those after OSS. At 3000 µmol/L of sulfate, calcium binding increases so that approximately 13% of calcium is bound. This leads to significant hypocalcemia and increased anion gap,[8] neither of which occurred after OSS. Unlike the reduction in GFR that follows oral sodium phosphates for bowel cleansing,[9] renal function parameters were unchanged after OSS. These observations provided support for an enlarged clinical program with OSS.[7]

Limitations of this study include its small size and inclusion of subjects who were younger than those who usually undergo colonoscopy. However, large phase III studies with this OSS found no evidence of an effect of OSS on markers of renal or hepatic function.[7] In other studies in rats and dogs, OSS at 7 times the human dose on a body weight basis for 30 days had no significant effect on renal or hepatic markers (data on file; Braintree Laboratories Inc). This suggests that this sulfate formulation can be safely administered to patients with moderate renal or hepatic impairment. Plasma sulfate levels would be expected to be higher in patients with more severe renal disease.

This study probably overestimated the apparent fraction of the sulfate dose that appeared in urine. Attempts to correct the urine sulfate concentrations during the dosing interval for the predose levels gave negative values in many instances (data not shown), suggesting that in these fasting individuals, endogenous sulfate excretion may have been diminished. In any event, because renal and hepatic diseases are not known to alter sulfate absorption, health status did not apparently alter either the amount of sulfate excreted or absorbed.

The urinary excretion of endogenous substances may more reliably reflect their absorption than the AUC.[10] Applying this approach to the urinary data buttresses the conclusion that the absorption of sulfates in this study was unaffected by moderate renal or mild-moderate hepatic impairment and that the estimate of the absorbed sulfate fraction is close to the amount excreted in the urine.

These observations, in addition to the lack of changes in key electrolyte values, warrant the further exploration of the safety and efficacy of this sulfate formulation for bowel cleansing in these patient groups.

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